Sri Devaki Meduri

ep23btech11030@iith.ac.in
https://www.linkedin.com/in/sri-devaki-meduri-43b5953a1/ Phone: +91 97898 41633

Curriculum Vitae

Curious and passionate about Research in Astronomy, Astrophysics and Cosmology. Working actively on Cosmology, Astronomy and Astro-particle Physics projects. Enthusiastic to dive deeper into other branches of Astrophysics with the guidance of expert professors and guides.

Education

- **Indian Institute of Technology, Hyderabad** (2023 - 2027)
  Bachelor of Technology in Engineering Physics with Minor in Mathematics and Honors in Physics
  CGPA: 9.51 / 10
- **Padma Seshadri Bala Bhavan Senior Secondary School, Siruseri** (2021 - 2023)
  12th Grade (2022-23) Percentage (CBSE): 97.8% (489/500)
- **Padma Seshadri Bala Bhavan Senior Secondary School, Siruseri** (2011 - 2021)
  10th Grade (2020-21) Percentage (CBSE): 98.2% (491/500)

Research Interests

Cosmology; Gravitational Waves; Black Holes; Dark Matter and Dark Energy; Neutron Stars; Astro-particle Physics; AI and Data Science

Publications and Preprints

- **Cosmology with Gravitational Wave Sources**
  (Project as a part of NIUS (National Initiative on Undergraduate Science *(refer below)*) under Prof. Suvodip Mukherjee of TIFR, Mumbai)
  Working on black hole mass distribution using gravitational waves. Paper being prepared for submission.
  Brief Idea of the Project The project involves generating data using software like GWSim and analysing the data to understand the mass distribution.
- **Limit on high energy neutrino emission from Abell 119 using IceCube 10-year muon track data**
  Project with Prof. Shantanu Desai of IIT Hyderabad. Paper under review in Astroparticle Physics (arXiv: 2605.11966).
  Brief Idea of the Project The project is primarily data analysis of IceCube data.
- **Test of ΛCDM using Galaxy-Galaxy lensing**
  Project with Prof. Shantanu Desai of IIT Hyderabad. Manuscript under review in Physics of Dark Universe.
  Brief Idea of the Project The project is primarily data analysis and model fitting of KiDS data. It compares several models and drawns conclusions.
- **Verification of BFJR**
  Project with Pradyumna Sadhu of UCR. Work in Progress. **Brief Idea of the Project** The project deals with verifying the BFJR relation with galaxies generated by the IllustrisTNG simulations.

Other Work

- **SPARC HSBs and LSBs**
  (B.Tech project with Prof. Shantanu Desai of IIT Hyderabad) Studying the SPARC dataset for understanding behaviour of Dark Matter Surface Density across galaxies of different surface brightnesses.
  Brief Idea of the Project The project is primarily data analysis and model fitting of SPARC data.
- **National Initiative on Undergraduate Science (NIUS) Astronomy Camp**
  Astronomy camp that included several astronomy lectures along with a visit to Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune and Giant Metre-Wave Radio Telescope (GMRT), Narayangaon, Pune.
- **Teaching Assistant for Electrodynamics Course**
- **Teaching Assistant for Special Relativity Course**
- **Teaching Assistant for Electricity and Magnetism Course**
- **Teaching Assistant for Maths for Physics**
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- **Teaching Assistant for Introduction to Programming Course**
- **White Dwarfs and Neutron Stars**
  Reproducing the mass and density versus radius results for white dwarfs and neutron stars for various Equation of States.
- **Public Outreach Sessions (Part of Cepheid Club)**
  As a part of the Astronomy club of IITH, Cepheid, I have organised several stargazing sessions for general public. Also presented sessions on the Universe for school children.
- **Open Lectures to College Students (Part of Cepheid Club)**
  Conducted several sessions for college students (undergraduates, postgraduates and PhD students) Topics included The Hot Big Bang, Black Holes, and Gravitational Waves (included both theory and hands-on sessions).
- **Tuition for High School Students**
  Organized tuition for grade 11 and 12 students in Physics and Maths.
- **Fourier Theory**
  Extensive summary of Fourier Theory including Fourier transforms, Fourier series, Fourier transforms on distributions, Laplace transforms and z-transforms for College students.

Technical Skills

Programming Skills

- Python: Emcee, Astropy, Numpy, Matplotlib, PyCBC, Bilby, BBHxm
- GWSim: Software for gravitational wave sources manipulation (includes creation of simulated universe with galaxies, detection of black hole mergers for gravitational wave creation and detection of generated gravitational wave in desired detector)
- C
- Java Script and React
- SQL
- Solid Edge
- CAD

Other Skills

- SAOImage DS9: Software for visualizing .fits files
- Telescope Operation: both manual and automated

Achievements and Extra-curriculars

- Academic Excellence Award for 2023 in IIT-Hyderabad, for having topped in the department with highest CGPA.
- VSSF SPOT 100 student. Top 100 in India in VSSF SPOT Quiz. Taken to Satish Dhawan Space Centre, Sriharikota (SDSC-SHAR) in 2019.
- Domain Head of Astrotech domain of Cepheid, the Astronomy Club of IITH.